

University of Moratuwa

ETHICS APPLICATION - FORM B FOR ABOVE-LOW RISK RESEARCH

In order to find out whether you do require to submit the human ethics application form B please read the **general guidelines page**.

Incomplete or inadequate applications or applications prepared on out-dated application forms will not be considered by the ethics committee.

Submission Details:

University Ethics applications: The completed application should be submitted via e-mail to ar-fgs@uom.lk in the form of a **pdf document** either scanned with signature(s) or signed electronically of the application plus all attached documents, **also in pdf format**.

University of Moratuwa respects the privacy of your personal information. We collect personal information about you on this form to review and process your human research ethics application with respect to the University Ethics Committee's responsibilities. For this purpose, we will use this information and typically disclose it to funding or regulatory bodies pertaining to the research under consideration. Information provided in this application will be treated confidentially.

YOU ARE REMINDED THAT NO RESEARCH STUDY (DATA COLLECTION OR ANALYSIS OF SECONDARY DATA) SHOULD BE CONDUCTED WITHOUT THE PRIOR ERC APPROVAL

University of Moratuwa

ETHICS APPLICATION - FORM B FOR ABOVE-LOW-RISK RESEARCH

1. Project Title:	Beyond Optical PPG: mmWave Radar Arterial Pulse Sensing	
2. Chief Investigator / Supervisor: (academic staff members only)	Name: Dr. Sampath Perera Position: Senior Lecturer Department: Department of Electronic and Telecommunication Engineering	
Student (if appropriate)	Name: Thamilezai Ananthakumar Kopithan Manoranjithan Mathujan Shanmugavadivel Sivamynthan Nadesamoorthy Course of Study: Final Year Undergraduate Project (B.Sc. Eng.) Department: Department of Electronic and Telecommunication Engineering	
3. Project Duration: (subject to annual review)	Project commences: July 2026	Project concludes: May 2027

Prof. Kandappu is an external advisor and collaborator on this project. She will provide academic guidance and will not have access to participant data, identifiable information, or the research datasets collected during the study. Access to the collected research data will be restricted to the investigators listed in this application. Any data shared publicly through journal publications or public datasets will only be made available in accordance with the approved data-sharing and anonymization procedures

4 PROJECT

- (a) **Please provide a two sentence lay summary of the project** (50 words or less. Please ensure the description is in plain language to prevent the return of the application)
- This project aims to develop an adaptive non-invasive heart rate and blood pressure monitoring system using mmWave radar, providing a more reliable alternative to conventional optical (PPG) sensors for cardiovascular monitoring. The proposed technology is designed to overcome the limitations of conventional optical sensors, including ambient light interference, variations in skin pigmentation, and reduced measurement accuracy caused by differences in subcutaneous fat thickness.
- (b) Please attach your **complete research proposal**

The complete research proposal is attached, along with all the necessary supporting documents.

5 PERIOD DURING WHICH ACTIVITIES REQUIRING ETHICS APPROVAL WILL OCCUR
From: October 2026**To:** April 2027**Indicate what areas in the project require ERC approval:**

Data collection from participants using non-invasive radar and photoplethysmography (PPG) measurements.

Collection and storage of basic participant demographic data.

Participant recruitment and informed consent procedures.

6 FUNDING (tick whichever is applicable)

- (a) ☒ Will not be sought
☐ Will be sought
☐ Has been sought
☐ Has been obtained

Name of funding agency:**Project title under which funding sought:****Please attach a copy of the funding application.**

- (b) **Will participants be informed of funding source(s)?**

☐ No (please explain)☐ Yes (provide details)☒ Not applicable**7 DESCRIPTION OF PROJECT**

Aims: Replacing issues in current optical-based sensing (ambient light, fat layer issues & colour pigmentation issues) with adaptive mmWave (radar) based PPG sensing

Participants:

Approximately 30 healthy adult volunteers ranging in age from 22 to 26 years will be recruited (university of moratuwa students). However, if sufficient variation in skin tone and BMI/subcutaneous fat thickness cannot be achieved within the UoM student population, participants may also be recruited from outside the University. The proposed sample size is consistent with prior studies in wearable and non-invasive physiological monitoring and is considered sufficient for system development and preliminary performance validation. Participation will be entirely voluntary, and written informed consent will be obtained prior to data collection.

Brief Description:

The study involves the simultaneous collection of physiological data using a custom wristband prototype and a Polar H10 chest strap and FDA-approved Omron wrist-cuff monitor device as the reference device for heart rate and blood pressure. The wristband integrates a 60 GHz mmWave radar, an optical photoplethysmography (PPG) sensor, and an inertial measurement unit (IMU). The objective is to compare the reliability and accuracy of radar-based and optical PPG-based heart rate monitoring under varying activity levels, including sitting, walking, and running. The study will also evaluate the effects of participant-specific factors, such as skin tone (skin pigmentation) and subcutaneous fat thickness, as well as physiological conditions such as sweating, on the performance of both sensing modalities.

Methods of Data Analysis:

Signals will be preprocessed (noise removal, filtering), features extracted (heart rate and waveform characteristics), and analyzed using Python algorithms. Results will be validated by comparing the wearable sensor outputs against the reference devices—the Polar H10 for heart rate and the Omron wrist cuff for blood pressure—using accuracy and reliability metrics.

8 PARTICIPANT DETAILS

(a) **Number of Participants:** 30

(b) **Age Range:** 22-26

(c) **Are there any criteria that will determine whether participants are included or excluded from the research?**

☐ **No**

☒ **Yes**

Inclusion Criteria: Healthy university students aged 22–26 years who are capable of providing informed consent and represent a range of skin pigmentation and subcutaneous fat thicknesses.

Exclusion Criteria: Individuals with diagnosed cardiovascular diseases or conditions affecting peripheral circulation

(d) **Recruitment Method**

Please state how names and contact details of potential participants will be obtained, from where they will be recruited, how they will be invited to participate, and who will approach potential participants to seek their participation.

Note: Where participants are recruited from schools, hospitals, prisons or other institutions, permission/approval from the institution or appropriate authority must be sought. See Question 11.

Participants will be recruited from the university community through convenience sampling. Interested volunteers will be screened for eligibility and provided with study information before participation.

(e) **Compensation to Participants**

☒ **Not applicable**

☐ **Applicable (provide details if any financial or other reward or compensation will be offered to participants)**

(f) **Involvement of Special Groups**

If the project involves groups requiring special permission (e.g. people such as children or the intellectually disabled who are in a dependent relationship requiring permission from a parent or guardian, describe the nature of the

group(s), justify the inclusion of people from the group(s), and specify the procedures to be used to obtain permission, and complete Question 12. (Please note that if participants are from a vulnerable group, their inclusion is justified only if the study cannot be done in any other group.)

☒ Not applicable

☐ Applicable (provide details)

(g) Are any of the participants Moratuwa University students?

☐ No

☒ Yes (Please explain the steps taken by the investigators to ensure that the subject's participation is purely voluntary)

Providing information sheet & Consent form

(h) Are any of the participants (other than Moratuwa University students) in a dependent relationship with any of the investigators (including those involved in recruiting, interviewing etc)?

☒ No

Participants recruited from outside the University of Moratuwa will not have any dependent relationship with the investigators. Participation will be entirely voluntary, and participants will be informed that they may decline or withdraw from the study at any time without any consequences.

☐ Yes (Please explain the relationship and the steps taken by the investigators to ensure that the subject's participation is purely voluntary)

9 RESEARCH USING EXISTING DATA BASES

Not applicable

If research involves access to existing data bases provided by an institution(s), please indicate:

(a) source(s) and number of records;

(b) whether data to be used will be de-identified, potentially identifiable (eg. coded), or identified; and

(c) whether permission has been granted by donors to use these data for research purposes.

(d) whether formal permission/clearance has been sought or obtained from the relevant institution(s) (see also Question 12 below).

10 LOCATION OF STUDY

Identify the location of the study, if other than Moratuwa University. If permission is required to use the location, indicate how permission will be obtained.

11 EXTERNAL APPROVALS

If a project requires approval from other institutions, next of kin or guardian, or representative or authority in the case of special groups, copies of such approvals must be provided to the UERC at the time of application or be made available as soon as possible thereafter.

Please indicate as appropriate if formal clearance/permission has been sought or obtained.

(a) Institutional

Name(s) of institution/ethics committee/authority:

☐ Yes, attached

☐ Yes, to follow (estimate when likely to be obtained):

☒ No (please explain)

The study will primarily be conducted within the University of Moratuwa premises with UoM student participants. However, if sufficient variation in skin tone and BMI/subcutaneous fat thickness cannot be achieved within the UoM student population, participants may also be recruited from outside the University. No external institutional approval is required.

(b) Next of Kin (or representative or authority in the case of special groups)

☐ Yes, attached

☐ Yes, to follow (estimate when likely to be obtained):

☒ No (please explain) Not applicable.

12 INFORMED CONSENT

(a) Indicate whether individual Information Statements/Information and Consent Forms will be used.

☒ Yes (please provide copies).

☐ Alternative method of obtaining consent will be used (please specify)

☐ No (please explain)

(b) Will the Information Statement/Information and Consent Form (ICF) be translated into the participants' first language?

(Note: if there is an ICF it MUST be translated. If the Informed consent is not being obtained, then the applicant has to make a strong justification.)

☒ Yes (please provide copies of translations)

☐ No (please explain)

- (c) **Will ALL participants have the capacity to give voluntary and informed consent?**

Yes ☒ [X]

No ☐ []

If No, who will be asked to provide consent?

☐ [] Parent/guardian

☐ [] Other - give details

(e.g. Normally a Legally Authorized Person as defined by Law)

How will consent be obtained?

☒ [X] Written consent form

☐ [] Verbal - explain below how consent will be recorded

(Note: Verbal is not acceptable unless there is an extremely unusual circumstances such as illiteracy, physical disability or other limiting condition)

13 DESCRIPTION OF PROCEDURES

- (a) **Describe in detail exactly what you will be asking of participants, and emphasise anything that may have adverse consequences.**

Each participant will take part in a single data collection session, expected to last approximately **60–90 minutes** in total, conducted in the following stages:

Stage 1: Briefing and Consent (~10 minutes)

The study procedure will be explained to the participant, and informed consent will be obtained. Participants will be given the opportunity to ask questions before proceeding.

Stage 2: Demographic and Anthropometric Data Collection (~10 minutes)

Basic demographic information (age, sex, height, weight) will be recorded. Skin tone will be assessed using a Fitzpatrick classification, and body fat classification will be done using the BMI.

Stage 3: Sensor Placement (~10 minutes)

The Polar H10 chest strap will be fitted around the participant's chest. The multimodal wristband prototype and the Omron BP4350 wrist-cuff monitor will be placed on opposite wrists, both positioned over the radial artery. Sensors will be checked for signal quality before recording begins.

Stage 4: Baseline Recording (~5–10 minutes)

Participants will remain seated at rest to allow for a stable baseline physiological recording, which will also serve as a reference condition for later comparison against motion conditions.

Stage 5: Activity Protocol (~20–30 minutes)

Participants will perform a sequence of physical activities while physiological signals are continuously recorded:

- Sitting (rest, extended duration)
- Walking (self-paced, on a flat indoor surface)
- Running (slow, on indoor track)

Short rest intervals (~2–3 minutes) will be provided between activities to allow heart rate to

stabilize and to reduce participant fatigue.

Stage 6: Sensor Removal and Debrief (~5 minutes)

All devices will be removed, and participants will be thanked and given the opportunity to ask any final questions.

No blood, tissue, or invasive samples will be collected at any stage. All procedures are non-invasive and use wearable, contact-based sensors only.

(b) If questionnaire(s) (including those that are published or commercially available) will be used in the project, please attach a copy to this application. (Please provide the questioner in a language in which it is to be administered. If it is in a language other than English please provide a translation

No questionnaires will be used in this project

(c) If interviews or focus groups are to be held, please attach the interview guide or focus group guide.

No interviews or focus groups will be conducted

(d) Will any form of deception, concealment or covert observation be used?

☒ Not applicable

☐ Details of deception, concealment or covert observation, justification for deception, debriefing procedures and debriefing documentation for participants attached.

14 SAFETY

The following check list has been developed to help identify specific safety issues:

YES NO N/A

- | | | | |
|-------------------------------------|-------------------------------------|--------------------------|---|
| ☒ | ☐ | ☐ | (a) Will participants come into contact with any equipment or apparatus? |
| ☒ | ☐ | ☐ | (b) Will participants come into contact with any equipment connected to or powered by an electricity supply in any form? |
| ☒ | ☐ | ☐ | (c) If the response to (a) or (b) above is yes, is the equipment standard, unmodified or commonly used, such as a computer or tape recorder? If no, please proceed with the questions below. |
| ☐ | ☒ | ☐ | (d) Will any equipment be used in any way that is different to its intended use or to the manufacturer's instructions? |
| ☒ | ☐ | ☐ | (e) Will participants be exposed to electrical or magnetic stimulation or non-ionizing radiation of any kind? |
| ☐ | ☒ | ☐ | (f) Will participants be exposed to external irradiation of ionizing radiation or be administered any radionuclides |
| ☐ | ☒ | ☐ | (g) Will participants be exposed to noise levels exceeding a peak sound level of 140dB or an 8 hour sound level of 85 dB equivalent? |
| ☐ | ☒ | ☐ | (h) Will participants come into contact with or be required to operate or use plant or machinery? |

If the response to "Yes" is ticked, please provide full details of the equipment, venue and procedures to be used. Provide details of all safety and protective measures to be in place if needed, evidence that your equipment has passed relevant electrical or other safety assessments. In addition, research projects involving the use of ionising radiation require approval from a relevant Radiation Safety Body or Health Department). Please attach evidence that such approvals have been obtained or have been sought.

Participants will come into contact with a wearable prototype containing an ams-osram optical front-end and an Infineon 60 GHz radar sensor and an inertial measurement unit (IMU). In addition, participants will wear a Polar H10 chest strap and FDA-approved Omron wrist-cuff monitor, which will serve as the reference device for heart rate and blood pressure measurement. These sensing components and reference devices have been widely adopted in previous physiological monitoring studies for wearable prototype development, radar-based vital sign sensing, optical PPG acquisition, and reference heart rate and blood pressure data collection. The radar transmits a low-power electromagnetic wave toward the tissue to measure mechanical deformation from arterial pulsation. This constitutes safe, low-power non-ionizing radiation that is standard for commercial near-field sensing. No electrical current is applied to the participant. The study involves only non-invasive measurements and is considered minimal risk.

15 POTENTIAL BENEFITS

(a) To the participant

Participants may not receive direct medical benefits but may gain awareness of their physiological measurements recorded during the study.

(b) To humanity generally

The study contributes to developing robust, non-invasive cardiovascular monitoring systems that reduce limitations of conventional optical sensors, such as ambient light interference, skin pigmentation, and tissue variation effects.

16 RECORDING AND SECURITY OF PROJECT DOCUMENTATION

(a) How will data be recorded? (eg. written questionnaires, interview notes, photographs, audio/video recording, direct electronic data entry)

Password-protected laptop, access restricted to investigators

(b) Will confidentiality of results be maintained?

☒ Yes (detail below)

All electronic data files will be labeled using anonymous identification codes rather than personal identifiers. Consent forms will be stored separately from the research data.

☐ No (please explain)

(c) Indicate how the security of project documentation will be maintained and specify the precise location of the storage place(s):

(normally 5 years for non clinical trial data and 15 years for clinical trial data following publication).

(i) **during the study** : Paper documents (consent forms) will be stored in a locked cabinet within the Department of Electronic and Telecommunication Engineering. Electronic data will be stored on password-protected computers accessible only to the research team.

(ii) **following completion of the study**: All study records will be retained securely on password-protected systems and in locked cabinets for a minimum of five years, after which they will be destroyed.

(d) Will data (including samples) be preserved for possible future use in another project?

☐ No

☒ Yes (please explain the nature of the data to be preserved, when the data might be used in another project, how that data might be used, for what purpose it might be used, and who might be given access to the data for another project).

Data may be preserved for future extensions of the same research project or related studies, only with additional participant consent. The preserved data will be used for further analysis, validation, and improvement of the proposed cardiovascular monitoring system. Access to the data will be limited to the research team and approved collaborators.

17 DISSEMINATION OF RESULTS

(a) Will participants be informed that results from the study may appear in publications, be included in a thesis or report, or be presented at conferences? (if relevant, this information should be included in the Information Sheet given to participants prior to obtaining informed consent).

☒ Yes (give details)

Study results may be included in publications, thesis reports, or conference presentations. Data will only be presented in anonymized or aggregated form to protect participant confidentiality.

☐ No (please explain)

(b) Will participants be informed that results from the study will be available to them on request? (If relevant, this information should be included in the Information Sheet given to participants prior to obtaining informed consent).

☒ Yes

☐ No (please explain)

Will participants be informed that their personal data collected in the course of the research will be available to them on request? (if relevant, this information should be included in the Information Sheet given to participants prior to obtaining informed consent).

☒ Yes

☐ No (please explain)

18 . Identify any other potential ethical issues not listed above.

No additional ethical issues have been identified. The study relies entirely on non-invasive sensors to assess cardiovascular dynamics. The radar system uses low-power, non-ionizing electromagnetic radiation, and the sensing components employed in the study have been previously used in research applications. The potential risk to participants is minimal, with only a possibility of minor sensor heating during prolonged use, which will be monitored to ensure participant comfort and safety. Participation is voluntary, and participants may withdraw from the study at any time.

19. Investigators Working on Procedures that Involve humans:

By signing each investigator acknowledges that the information provided is true and they are aware of the procedures they are to perform.

Chief Investigator/ Principal Supervisor (University of Moratuwa academic staff only)

Name:

Signature:

Date

Student/ Investigator 2

Name: Mathujan.S

Signature:

Date: 24.08.2026

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- ☐ Clearance from a medical faculty not required
- ☐ Clearance from the Faculty of Medicine, University of has been obtained. Approval is attached herewith.
- ☐ Clearance from the Faculty of Medicine, University of, if the research involves animals, has been obtained. Approval is attached herewith.

Ethics Approval Number:

Chairperson of UERC

.....		
Name	Signature	Date

19 ATTACHMENT CHECKLIST

The following documents are attached (please tick where appropriate):

- ☒ copy of the research proposal
- ☐ copy of funding application (s)
- ☐ evidence of permission to use places off-University
- ☐ evidence of external approvals (Questions 11(a) and (b))
- ☒ copy of the proposed Information Sheet(s)
- ☒ copy of the proposed Consent Form(s)
- ☒ copy of certified translations of Information Sheet(s) and Consent Form(s)
- ☐ copy of questionnaire(s) and/or proposed interview/focus group outline
- ☐ copy of relevant safety approvals